1. PROGRESS UP TO DATE

1.1 Signal Model and Orthogonality Principle

In the suggested multi-user model, complex (analytic) signal representation of the trigonometric chirp signal in each user is represented as given in equation (1.1).

$$s_u(t) = b_u\sqrt{P}e^{j(\omega_c t+\beta sin(\omega_u t))}u = 1,2,3 \dots, U \tag{1.1}$$

In this case, the message bit sent by the u th user is $b_u \in \{-1 + 1\}, \omega_c$ carrier frequency, β modulation index, P transmission power (all users have the equal transmission power) and ω_u baseband frequency of the $u - th$ user. Considering a bit duration of T the waveform of the real form is represented as given in equation (1.2).

$$s_u(t) = Re\{s_u(t)\} = \sqrt{2P}cos\big(\omega_c t + \beta sin(\omega_u t)\big), \qquad 0 \le t \le T \tag{1.2}$$

The total signal $s_T(t)$ reaching the receiver from U users is given in equation (1.3).

$$s_T(t) = \sum_{u=1}^{U} s_u(t) \tag{1.3}$$

In the transmitter section of the proposed multi-access system, the trigonometric chirp signal carrying the bit information b_u for the $u - th$ user can be expressed in terms of Bessel functions $J_0(\beta)$ as in equation (1.4), representing the harmonic components located at the frequency $\omega_c + n\omega_u$.

$$s_u(t) = b_u \sum_{n=-\infty}^{\infty} J_n(\beta)e^{j(\omega_c+n\omega_u)t}\, u = 1,2,3 \dots, U \tag{1.4}$$

The amplitude value (at the harmonic frequency as in **Figure 1.1** of the FM signal of the trigonometric chirp depends on the value of the modulation index of the chosen index of the modulations $J_0(\beta)$, according

to the chosen value of the modulation index (β). The orthogonality of the signal that is transmitted to each user in the proposed initiative is expressed in equation (1.5).

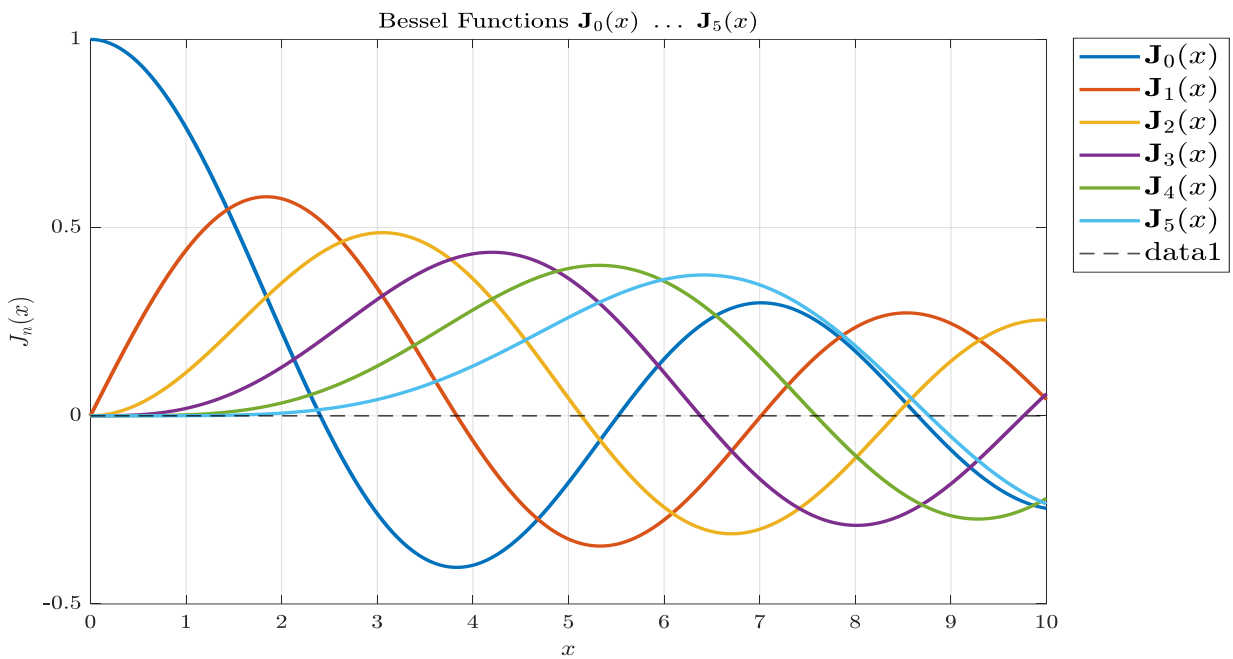

Figure 1.1:Variation of Bessel functions of the first kind, $J_n\,(x)$, from the zeroth to the fifth order.

$$\int_0^T s_k(t)s_l(t)dt, = \begin{cases} P & ,k = l \\ 0 & ,k \neq l \end{cases} \tag{1.5}$$

In order to meet this requirement, the carrier frequency component has no power, as is typical among all users, and is associated with the case where again, $|J_0(\beta)\,| \approx 0$. The values of the modulation index that fulfill this requirement are shown in **Figure 1.2**

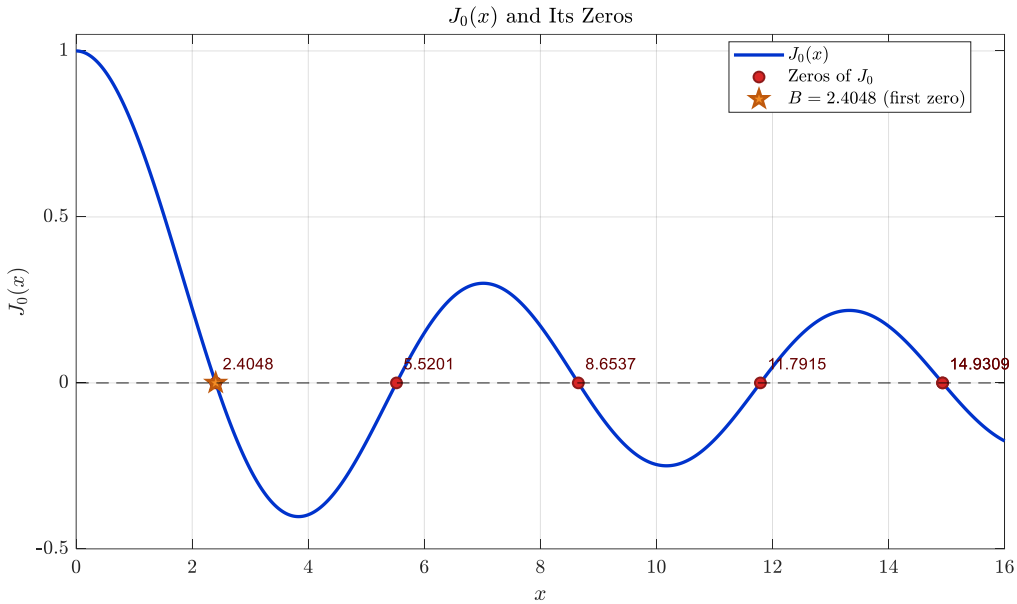

Figure 1.2: Plot of the Bessel function $J_0(x)$ and the β values satisfying the condition $|J_0(\beta)| \approx 0$.

1.2Frequency Allocation and Time–Frequency Characteristics

In this case, at the point where $J_0(\beta) = 0$ is zero, the carrier term, $(n = m = 0)$ will be zero and the common signal component will be entirely suppressed. Assuming that $f_1$represents the baseband frequency of the first user , the successive frequency values assigned to U users are given in equation (1.6).

$$f_{u+1} = f_u + \Delta f, u = 1,2 \dots U - 1 \tag{1.6}$$

In order to conduct a detailed analysis of the spectral efficiency and bandwidth requirements of the proposed multi-user system, the behavior of the signal in the time-frequency plane has been investigated. Based on the phase argument $\phi_u(t) = \beta \sin(\omega_u t)$ given in Equation (3.1), the Instantaneous Frequency of the $u - th$ user is derived via the time derivative of the phase, as shown in equation (1.7).

$$f_{i,u}(t) = \frac{1}{2\pi}\frac{d\phi_u(t)}{dt} = \frac{1}{2\pi}\frac{d}{dt}(\beta \sin(2\pi f_u t)) = \beta f_u \cos(2\pi f_u t) \tag{1.7}$$

Equation (1.7) reveals that the frequency trajectory of the proposed modulation technique exhibits a cosinusoidal structure with an amplitude of $\Delta f_{peak} = \beta f_u$ and an oscillation rate of f_u . This analysis demonstrates that as the user index u (and consequently the baseband frequency f_u) increases, both the rate of frequency change and the frequency deviation of the signal increase linearly.

To ensure these signals are transmitted without spectral overlap (aliasing), the effective bandwidth (BW_{eff}) is calculated as in equation (1.8) using Carson's Rule.

$$BW_{eff} \approx 2\left(\Delta f_{peak} + f_m\right) = 2(\beta f_u + f_u) = 2(\beta + 1)f_u \tag{1.8}$$

To provide maximum power efficiency of this study, the modulation index (β) is chosen such that it will satisfy the conditions shown in **Figure 1.2** at its minimum value, which represents the first root to ensure carrier suppression.When this value is used to substitute in Equation (1.8), the bandwidth of each individual user is found to BW_{eff} =6.81f_u. The normalized time-frequency analysis of the time-frequency plot for the system with five users is presented in **Figure 1.3**.

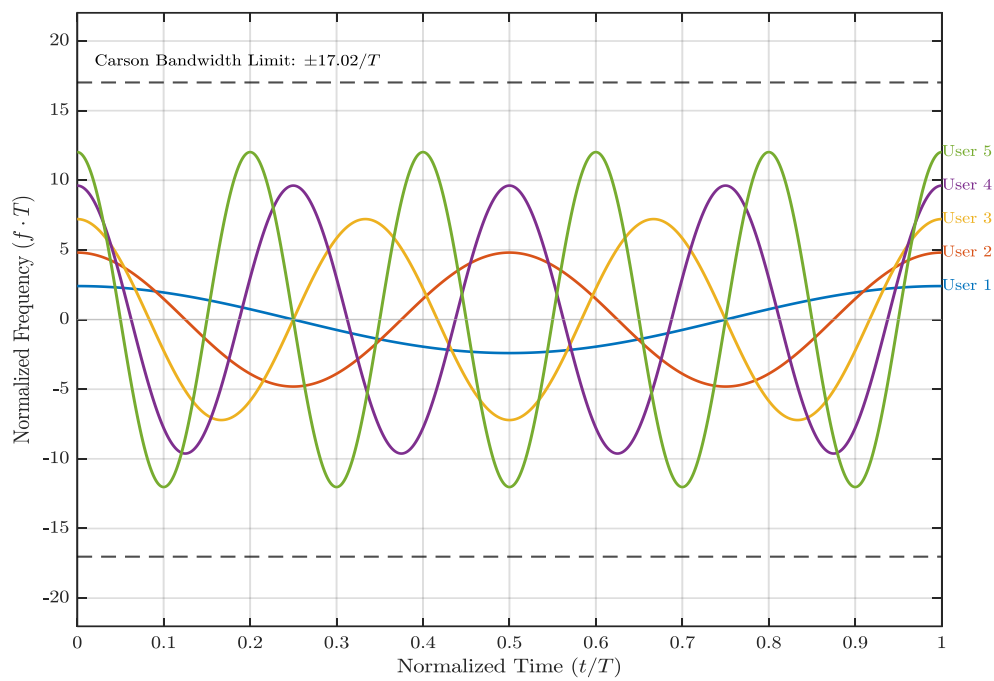

Figure 1.3: Normalized Time-Frequency representation for $K = 5$ users with the optimized modulation index β = 2.4048.

1.3 Harmonic Overlap Constraint and Spectral Analysis

To ensure mutual orthogonality among users in the spectrum, there must be no overlap in the allocated frequencies up to the harmonic component order $n > \beta + 1$ which accounts for the total signal power. The constraint governing this frequency allocation is provided in equation (1.9).

$$mf_k \neq nf_l, \qquad if \quad \min(m,n) \leq \beta + 1, \qquad k \neq l, \qquad k = 1,2,\dots,U, \qquad l = 1,2,\dots,U \quad (1.9)$$

As an illustrative example with $\Delta f = 1kHz$, $\beta = 2.4048$, negligible power is observed in the four-th harmonic ($n = 4$) and beyond, in accordance with the $n > \beta + 1$ criterion. In this simulation scenario, where $f_c = 100\ kHz$, $f_1 = 10\ kHz$ and U=5, the frequency spectrum of the composite signal $s_T(t)$,defined in (3) ,obtained at the receiver is presented in **Figure 1.4**. It is clearly evident from the figure that each user maintains orthogonality in the frequency domain and exhibits no mutual projection.

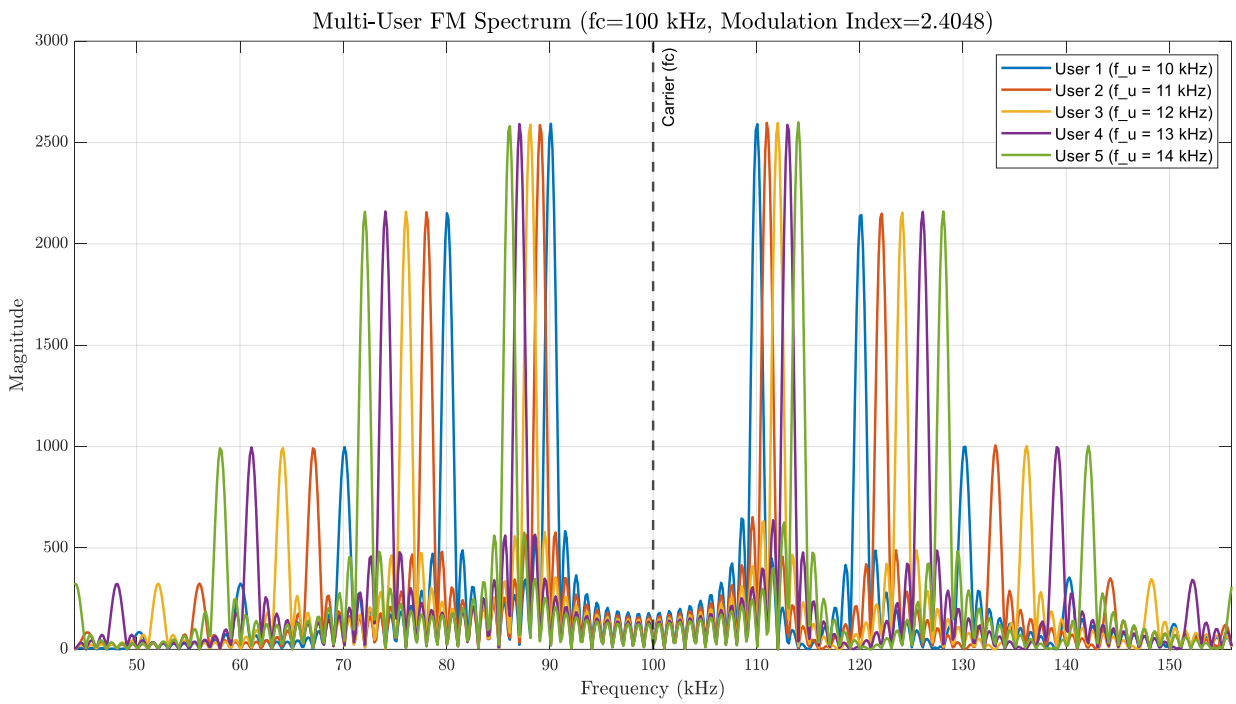

Figure 1.4: Spectral analysis of the proposed system for K=5 users. $f_c = 100\ kHz$, $\beta = 2.4048, \Delta f = 1kHz$.

According to **Figure 1.4**, the proposed system enables the achievement of low Multiple Access Interference (MAI) levels and high spectral efficiency without the need for additional windowing or filtering.

1.4 Cross-Correlation of Two FM Signals

Substituting the series expansions from equation (1.4) for both $S_1(t)$ and $S_2(t)$ into the integral in equation (1.2), the product of the two signals becomes the double sum given in equation (1.10).

$$S_1(t)\,S_2(t) = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} J_n(\beta)\, J_m(\beta)\, \cos(\alpha t)\, \cos(\gamma t) \tag{1.10}$$

where $\alpha = \omega c + n\omega_1$ and $\gamma = \omega c + m\omega_2$ denote the harmonic frequencies of the respective users. Applying the product-to-sum trigonometric identity to equation (1.10) yields equation (1.11).

$$\cos(\alpha t)\, \cos(\gamma t) = \frac{\cos\big((\alpha-\gamma)t\big) + \cos\big((\alpha+\gamma)t\big)}{2} \tag{1.11}$$

When integrated over $[-T/2, T/2]$, the high-frequency term at $(\alpha+\gamma)$ averages to zero for large T. Using the general antiderivative identity $\int\cos(kt)dt = (1/k)\sin(kt)$, the low-frequency term evaluates as given in equation (1.12) for the case $\alpha \neq \gamma$.

$$\frac{1}{2}\int_{-T/2}^{T/2} \frac{T}{2}\cos\big((\alpha-\gamma)t\big)\, dt = \frac{\sin\big((\alpha-\gamma)T/2\big)}{\alpha-\gamma} = sinc\big((\alpha-\gamma)T/2\big)\,\frac{T}{2} \tag{1.12}$$

Substituting $\alpha - \gamma = n\omega_1 - m\omega_2$ into equation (1.12) and summing over all harmonic pairs, the zero-lag cross-correlation is obtained as given in equation (1.13), valid for $\alpha \neq \gamma$.

$$R_{12}(0) = \frac{T}{2}\sum_n \sum_m J_n(\beta) J_m(\beta)\; sinc\big((n\omega_1 - m\omega_2)T/2\big) \tag{1.13}$$

Equation (1.13) shows that the cross-correlation is a doubly-infinite, Bessel-weighted sum of sinc functions. The sinc factor acts as a frequency selector: it approaches unity when $n\omega_1 \approx m\omega_2$ and decays as the harmonic frequency difference grows, directly quantifying the spectral overlap between user signals.

1.6 Auto-Correlation and Signal Power

For the special case where both signals are identical (i.e., $S_1 = S_2$, $\alpha = \gamma$, $n = m$), we have $\cos(0) = 1$ and $\alpha + \gamma = 2\alpha$. The auto-correlation integral then evaluates as given in equation (1.14).

$$R_{11}(0) = \frac{1}{2}\int_{-T/2}^{T/2} [1 + \cos(2\alpha t)]dt = \frac{1}{2}\left[T + \frac{\sin(\alpha T)}{\alpha}\right] \tag{1.14}$$

The oscillatory term $\sin(\alpha T)/\alpha$ becomes negligible for large T or sufficiently large α, representing the high-frequency residual. Discarding this term, the auto-correlation simplifies to equation (1.15).

$$R_{11}(0) \approx \frac{T}{2} \tag{1.15}$$

This result is consistent with the expected average power of a unit-amplitude cosine signal. By symmetry, the same relation holds for $R_{22}(0) = T/2$.

Substituting the results of equations (1.13) and (1.15) into the normalization formula and noting that $\sqrt{(R_{11}(0) \cdot R_{22}(0))} = \sqrt{(T/2 \cdot T/2)} = T/2$, the normalized cross-correlation coefficient is obtained as given in equation (1.16).

$$\rho_{12}(0) = \sum_{n} \sum_{m} J_n(\beta) J_m(\beta)\ sinc\big((n\omega_1 - m\omega_2)T/2\big) \tag{1.16}$$

Equation (1.16) expresses the similarity between two FM signals as a doubly-infinite, Bessel-weighted sum of sinc functions. When the two signals are identical ($\omega_1 = \omega_2$), the arguments of all sinc terms vanish, yielding sinc(0) = 1, and the Parseval-like completeness relation of the Bessel coefficients gives $\rho_{12} = 1$, confirming unit self-correlation. Conversely, when the baseband frequencies are sufficiently separated, the sinc terms decay toward zero and $\rho_{12} \to 0$, demonstrating signal orthogonality.

It is important to note that the expression in equation (1.16) is derived under the condition $\alpha \neq \gamma$, i.e., $n\omega_1 \neq m\omega_2$. Terms for which this equality holds contribute a constant offset and are treated separately as part of the auto-correlation, as shown in Section 1.7

1.7 Summary of Key Results

The principal results derived in this section are consolidated in Table 1.

Table 1 Summary of derived cross-correlation and auto-correlation expressions.

Quantity	Expression
Cross-Correlation $R_{12}(0)$	$\frac{T}{2} \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} J_n(\beta)\ J_m(\beta)\ sin\,c\left((n\omega_1 - m\omega_2)\frac{T}{2}\right)$
Auto-Correlation $R_{11}(0)$	$\frac{T}{2}$
Normalized Coeff. $\rho_{12}(0)$	$\sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} J_n(\beta)\ J_m(\beta)\ sin\,c\left((n\omega_1 - m\omega_2)\frac{T}{2}\right)$